

ZOMATO DELIVERY TIME ANALYSIS

This Is An Interactive Dashboard Where It Is Helps In The Analyzing The Problem Of The Time Management Of The Delivery Person And Also The Factors Driving The Delay Of The Delivery

Business Problem

The Company Is Having Raw Dataset , Which Consist Of Delivery Person Id, Delivery Person Age, Delivery Person Ratings, Restaurant Latitude, Restaurant Longitude, Delivery Location Latitude, Delivery Location Longitude, Order Date, Weather Conditions, Road Traffic Density, Vehicle Condition, Type Of Order, Type Of Vehicle, Multiple Deliveries, Festival, City , Time Taken (Min)

This Uncleared Data Is Not Enough For Deriving The Business Problem

So The Main Goal Of The This Project Is To Clean The Dataset And Derive The Business Problem The Company Is Facing

Objective

- Provide The Answer To The Business Question
- Finding What Factors are Affecting The Delivery Time
- Finding The Which Of Combination Of Weather And Traffic Is Affecting The Delivery Time
- KPI Keys Like Average Time

Dataset

Column	Description
Order id	Order unique id
Delivery person ID	Delivery person unique id
Delivery person Age	Age of that person
Delivery person Ratings	Rating of that person

Restaurant latitude	Restaurant Location details
Restaurant longitude	Restaurant Location details
Delivery location latitude	Destination location details
Delivery location longitude	Destination location details
order date	Date of order placed
Time Ordered	The time when order placed
Time order picked	The time when order is pick by delivery person
Weather conditions	What is the weather on the day of order
Road traffic density	Road clearance
Vehicle condition	Condition of the vehicle
Type of order	What type of order it is
Type of vehicle	What type of vehicle
multiple deliveries	Number of items per order
Festival	Is there any festival on the order day
City	What type of city It is
Time taken (min)	How much time taken for the delivery

Data cleaning

- Remove the records consisting of 0 and negative value in the location information.
- And removed the time ordered and time order picked because the time taken data is already given.
- Remove the records in the city and weather condition which has NAN value.
- Checked for the duplicate records.
- Change the text dates in to proper date format.
- Checked for the outliers in the delivery person age and delivery person rating column.
- Checked all data types.

Data analysis

Business questions

The main question in this project is which factors are affecting the delivery time , which is causing in the low customer retention.

KPIs (key performance indicators)

- Average age

Dashboard

KPI cards

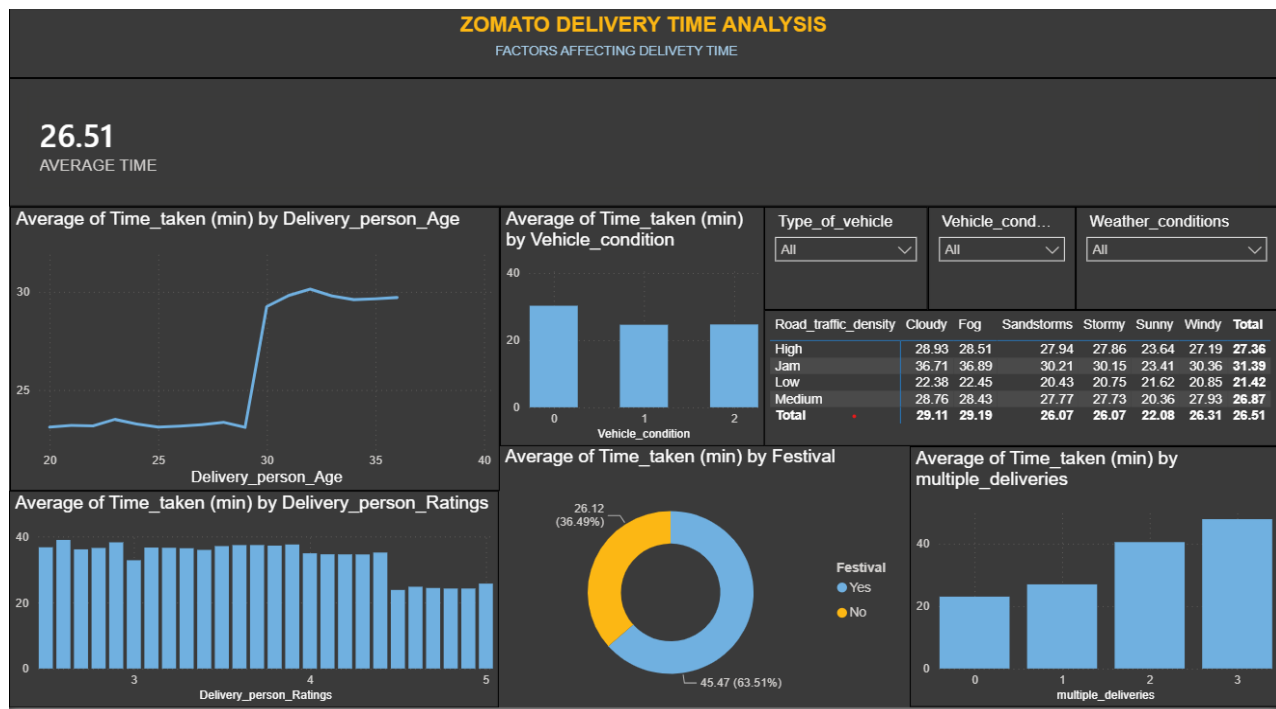
- Average age

Visualization

- Average of Time taken (min) by Delivery person Age
- Average of Time taken (min) by Delivery person Ratings
- Average of Time taken (min) by Vehicle condition
- Average of Time taken (min) by Festival
- Average of Time taken (min) by multiple deliveries
- Road traffic density, Time taken and Weather condition Matric

Filters

- Type of vehicle
- Vehicle condition
- Weather condition



Insights

1. The weather and road traffic contribute to the slow delivery

- The combination of the Jam and all weather as the average time of **31.39 min**, Which is more compared to other traffic conditions
- Mainly Jam and cloudy , Jam and fog this combination as the high delivery time records
- **Jam and fog ->36.89 min**
- **Jam and cloudy -> 36.71 min**

2. The vehicle condition also affecting the delivery time

The major down fall comes from the vehicle condition 0

- Vehicle condition 0 -> 38.23% (average time 30.25 min)
- Vehicle condition 1 ->30.84% (average time 24.58 min)
- Vehicle condition 2 ->30.91% (average time 24.67 min)
- Drivers operating poorly maintained vehicles (Score 0) experience a **23% speed penalty** compared to those on modern or well-maintained fleets."

We can see that vehicle condition 0 is causing 38.23% of slow delivery time

3. Festival days also affecting the delivery timing

- Festival time the delivery time increased by -> 63.51% (average time 45.47 min)
- Non Festival time the delivery time ->36.49% (average time 26.12 min)
- Festival days cause an operational crisis, slowing our network down by **74%** (jumping from 26 to 45 minutes).

4. The trip stacking also affecting the delivery time

- 3 deliveries ->34.59%
- 2 deliveries ->29.27%
- 1 deliveries ->26.96%
- 0 deliveries ->23.03%
- When we stack 3 deliveries on a single driver, that time skyrockets to **48.22 minutes**—a massive **108% increase** in customer wait time.

As the trip stack increase the time taken to delivery also increasing

Tools

Excel -> for cleaning the data

Power bi -> creating visualization and getting meaning full insights

Conclusion

This project gives the insights about the slow delivery time causes, and based on that the company should take action.

- **Reduce 3x Order Stacking:** We must update our driver app algorithm to cap stacked deliveries at a maximum of 2. Forcing 3 deliveries gridlocks our fulfillment.
- **Festival time Incentives:** Since festival days slow us down to 45 minutes, we need to introduce high-occupancy driver bonuses during

holidays to flood the streets with couriers and lower the load per driver.

- **Mandatory Vehicle Maintenance Audits:** Since Poor Vehicles (0) account for our slowest delivery brackets, we should implement a vehicle verification check in the app, prioritizing order assignments to drivers with a vehicle score of 1 or 2.